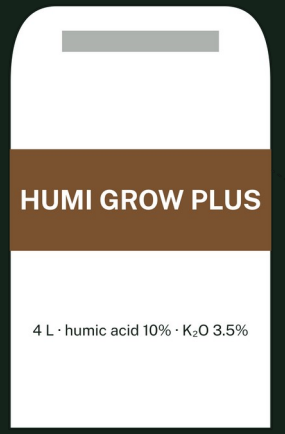
DEALER TRAINING · NOVEMBER 2026

Humi Grow Plus



Liquid potassium humate, humic acid 10% and K_2O 3.5%, 4 L, for the irrigation line at germination and early growth, made in Lahore.

VAN wrote this deck for its dealers and field team. A dealer can repeat its words to a grower.

THE 30-SECOND EXPLANATION

What a dealer says to a grower in 30 seconds.

Humi Grow Plus is potassium humate as a liquid: humic acid 10% and K_2O 3.5%, in a 4 L pack. It is the same active ingredient as the Humi Grow pellet, in a form that goes down an irrigation line. Humic substances add exchange sites to the soil, which hold potash, calcium, magnesium and ammonium for the root instead of letting them leach. They also bind (chelate) zinc, iron, manganese and copper, which keeps these micronutrients in solution at pH 8. In 81 published studies, humic substances increased root and shoot growth; this is research, not a result for this product. The liquid is applied at germination and early growth, while the root system is forming. Wheat takes 1 pack; maize and potato take 2. VAN's founder first made humic acid in Pakistan in 2007. Humi Grow Plus does not stop calcium from binding phosphate or change a soil test, and VAN gives no figure for phosphate saved.

The 4 points it must cover

1. The pack: humic acid 10%, K_2O 3.5%, 4 L, liquid, PS 5610-2023. Dilutes and goes down the line; no dissolving step.
2. The mechanism: exchange sites that hold nutrients, chelation that keeps micronutrients in solution, and root growth. It benefits the soil around the roots and the roots themselves.
3. The stage: germination and early growth, through the water, 1 to 2 packs an acre. It reaches the root while the root is forming.
4. The limits: it does not stop calcium from binding phosphate, VAN gives no figure for phosphate saved, and it does not change a soil test. The pellet form is Humi Grow, at land preparation.

Explain the product in simple words a grower uses, and keep to the figures on these slides. Every figure in this 30-second explanation is on the pack, in the published crop plan or in the named evidence on the slides that follow.

THE PACK

What the pack says, and what it is registered as.

4 L

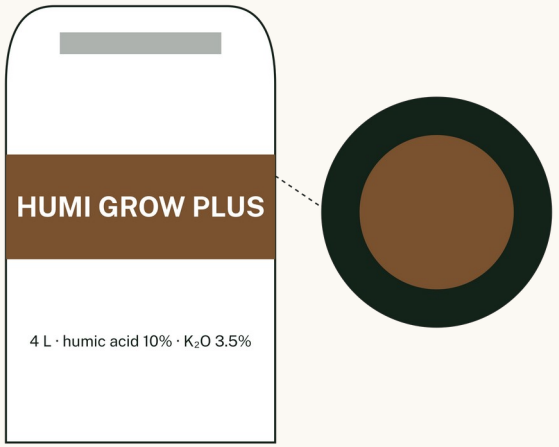
the pack. About 400 g of humic acid and 140 g of K_2O in it, reading the label percentages as weight per volume.

10%

humic acid, as the potassium salt, in solution. It is diluted in the irrigation water and carried to where the roots are.

K_2O 3.5%

the potash that comes with the humate. The crop plan counts it, but it is not the main source of potash.



Liquid

pours into the drum or the injector and goes down the line. No dissolving and no spreading.

2007

the year VAN's founder first made humic acid from Pakistani leonardite. By 2012 VAN was Pakistan's largest producer of humic acid.

0.61%

mean organic matter in 770,160 Punjab samples; 97.6% of samples below 1%. Most fields have little humic material left.

Registered as	Pakistan Standard	Manufacturing licence
Humi Grow Plus	PS 5610-2023	CM/L-3274/2024

If a grower asks about registration, give the licence number. Do not give an expiry date for the licence.

A GROWER'S QUESTION

“Everything on my potato goes through the drip now. Is there anything for the soil itself I can put through the line, or is the soil only for the spreader?”

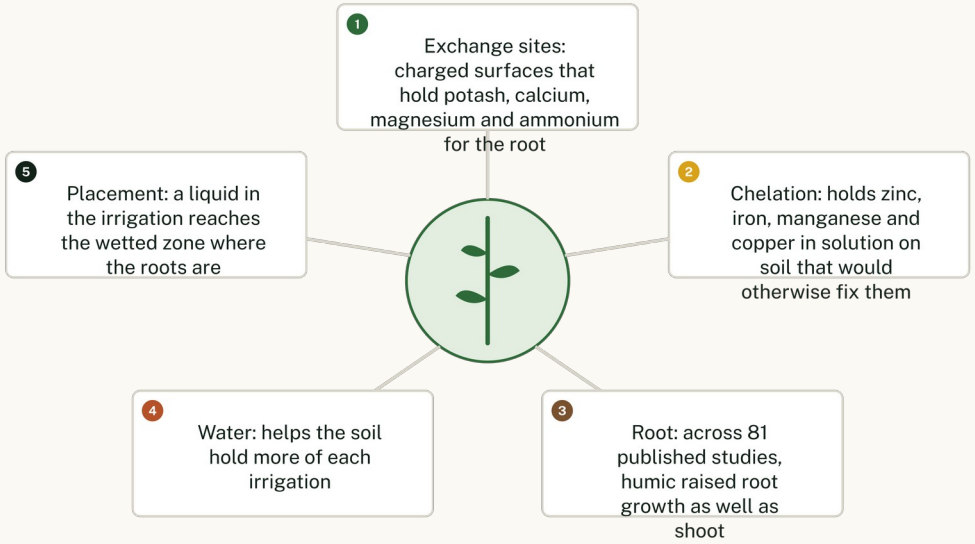
The grower now applies his nutrients through the water. The soil amendment he knows, the Humi Grow pellet, needs a drill or a spreader and is worked into the soil, so he thinks the soil cannot be treated through the line. The answer has 3 parts: what a humate does in soil, why the liquid goes on when the crop is already standing, and what it will not do.

The next 3 slides give the 3 answers, 1 idea on each slide.

ANSWER 1 OF 3

A humate adds sites in the soil that hold nutrients, and keeps micronutrients in solution.

Humic substances are the long-lived part of organic matter: large molecules covered in charged sites. Those sites hold potash, calcium, magnesium and ammonium where the root can reach them. The same molecules also bind zinc, iron, manganese and copper, a process called chelation, which keeps them in solution in soil at pH 8, where they would otherwise come out of solution. In 81 published studies, humic substances increased root growth and shoot growth (Rose et al. 2014); this is research, not a result for this product. van.com.pk states the benefit of VAN's humates in these terms: applied nutrients are held on the exchange sites instead of leaching or coming out of solution, and the root benefits.



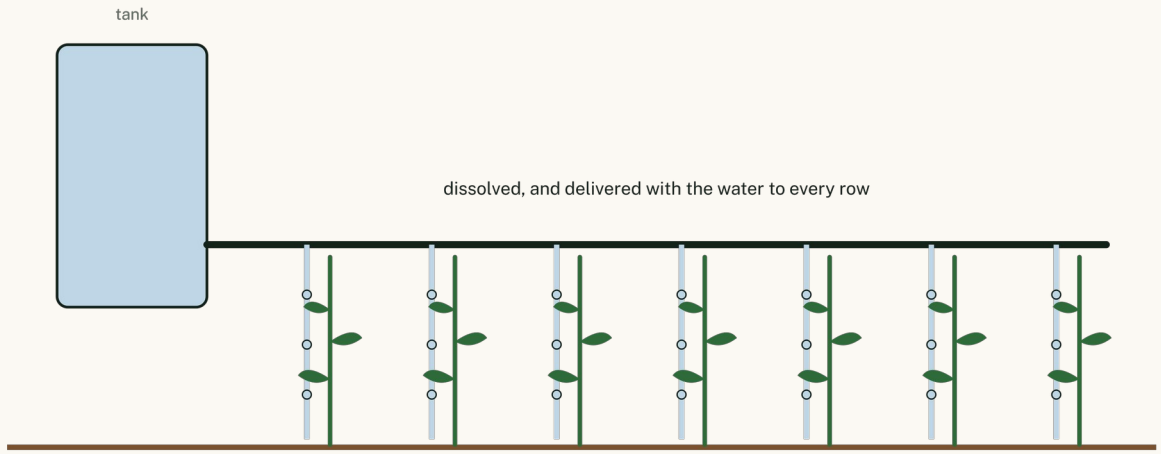
The 5 jobs of a humate, from textbook chemistry and the 81 published studies. The drawing illustrates the idea; it is not a measurement of this product.

Say it like this: "A humate helps your soil hold the nutrients you apply, and this is the form that goes down your drip line."

ANSWER 2 OF 3

The liquid reaches the wetted root zone after the crop is sown.

The pellet goes on at or before sowing, with the drill or the spreader, and is worked into the top layer of soil. A liquid injected with the irrigation goes where the water goes: into the wetted zone where the roots are, across the whole acre, with the crop already standing. That is why the crop plans apply Humi Grow Plus mostly at germination and early growth, while the root system is forming and the soil around it holds the nutrients the young root can reach. Wheat, garlic and citrus take 1 pack at germination; maize takes 2 at germination; potato takes 2 at early growth; strawberry takes 2 at early growth and 2 more at flowering; the sugarcane ratoon takes 2 at grand growth.



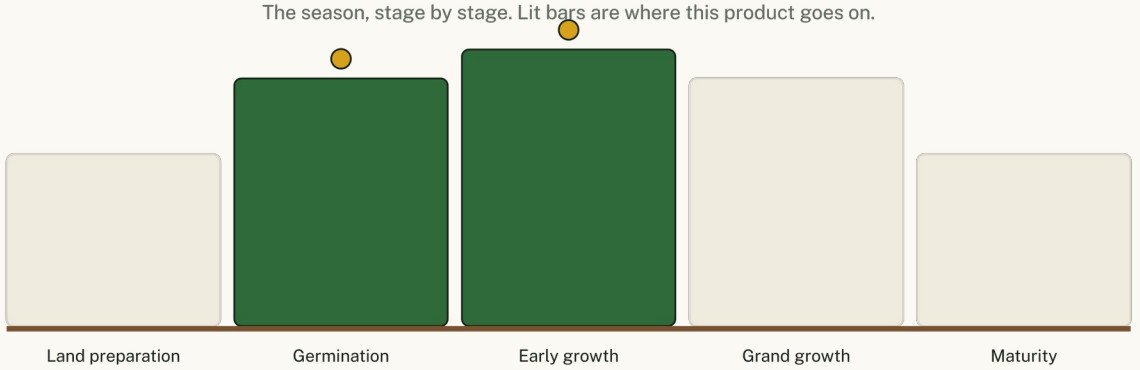
The fertigation line, from the drum at the head of the watercourse to the wetted root zone. The crop plan slides give the stage for each crop.

Say it like this: “Put it in the drum, with the water flowing, while the crop is young and its roots are forming.”

ANSWER 3 OF 3

What it will not do: stop calcium from binding phosphate, or change a soil test.

A humate is often sold as a product that stops calcium from binding phosphate. VAN tested that claim against its own soil data, and the claim does not hold: at the rates a grower would apply, humic acid binds far too little calcium to make a difference to the phosphate. So VAN does not make the claim, and gives no figure for phosphate saved. 1 pack puts about 400 g of humic acid (reading the label as w/v) on an acre, which is too little to change the organic-matter reading on the soil report. The effect is in the soil around the young roots. The 3.5% K_2O is 140 g a pack: it is counted, but it is not the main source of potash.



Germination and early growth are the stages the crop plan uses. The limits are stated because a dealer who overclaims for a humate loses the grower's trust when the next soil test does not change.

Say it like this: "It helps the root and the soil around it. It does not stop the lime in your soil from binding the phosphate from your DAP, and it will not change your soil test."

THE EVIDENCE

What is verified for Humi Grow Plus, and what is not.

No trial of Humi Grow Plus itself is on VAN's record. A dealer may state the following as facts, and should say where each comes from: what the pack declares, the licence, VAN's history with humic acid, the Punjab soil survey and the published research below.

Item	What it says	How a dealer may use it
The pack	Humi Grow Plus, liquid potassium humate, humic acid 10%, K_2O 3.5%, 4 L.	As fact.
The licence	Pakistan Standard PS 5610-2023, manufacturing licence CM/L-3274/2024.	As fact. The number, never an expiry date.
VAN's record	Humic acid made in Pakistan from Pakistani leonardite by VAN's founder in 2007; VAN the largest producer of humic acid in Pakistan by 2012; potassium humate made today in crystal, powder, pellet and liquid.	As fact, from the company timeline on van.com.pk.
The Punjab soil survey (770,160 samples, about 2016 to 2018)	Mean organic matter 0.61%; 97.6% of samples below 1%. 59.0% low in zinc; mean pH 8.22.	As fact about the soil. Never as product performance.
Published research, not VAN trials	Rose et al. 2014, a meta-analysis of 81 studies: humic substances raised shoot growth by about 22% and root growth by about 21%.	Spoken, as humic research. Never as a Humi Grow Plus result.
VAN's own check	VAN's 2026 calculation of how much calcium humic acid can bind: at field rates, humic acid binds too little calcium to stop calcium from binding phosphate. VAN does not make that claim.	As the reason the claim is never made.
Not on VAN's record	A Humi Grow Plus trial; a yield figure; a nutrient use efficiency percentage; a soil-test change; a percentage of phosphate or nitrogen saved; a foliar rate; a 20 L price.	Not said.

Sources: the Humi Grow Plus label; the regulatory table, the soil register and the company timeline on van.com.pk; the Punjab soil survey (770,160 samples, tested in about 2016 to 2018); Rose et al. 2014, Advances in Agronomy 124; VAN's 2026 calculation of calcium binding.

On wheat, 1 pack at Rs 1,240 per acre puts about 400 g of humic acid into the root zone.

The price is the delivered farmer price from the VAN farmer price sheet, August 2026. Humic acid and K_2O are from the declared analysis, read as weight per volume. Wheat takes 1 pack; maize and potato take 2; strawberry takes 4 over the season.

Humi Grow Plus, humic acid 10% · K_2O 3.5%	Volume	Humic acid	K_2O	Rupees
1 pack of 4 L	4 L	400 g	140 g	1,240
Wheat, garlic, citrus: 1 pack at germination	**4 L**	**400 g**	**140 g**	**1,240**
Maize at germination, potato at early growth: 2 packs	**8 L**	**800 g**	**280 g**	**2,480**
Strawberry: 2 packs at early growth, 2 at flowering	**16 L**	**1.6 kg**	**560 g**	**4,960**

On wheat, the grower spends Rs 1,240 per acre to put about 400 g of humic acid into the wetted root zone at germination. On potato he spends Rs 2,480 for 800 g at early growth. The 140 g of K_2O counts towards the crop's potash, but Humi Grow Plus is not the crop's main source of potash. VAN publishes no measured figure for the effect of Humi Grow Plus on the soil or the root.

Price: the VAN farmer price sheet, August 2026. 400 g = 4 L × 10% w/v; 140 g = 4 L × 3.5% w/v. Rates: VAN's published crop plans on van.com.pk.

The pellet and the liquid on the same crop plan: 2 forms at 2 stages, used together.

All 7 crop plans that name Humi Grow Plus also use the pellet; the table shows 5 of them. Rupees per acre are calculated from the delivered farmer prices in the VAN farmer price sheet, August 2026.

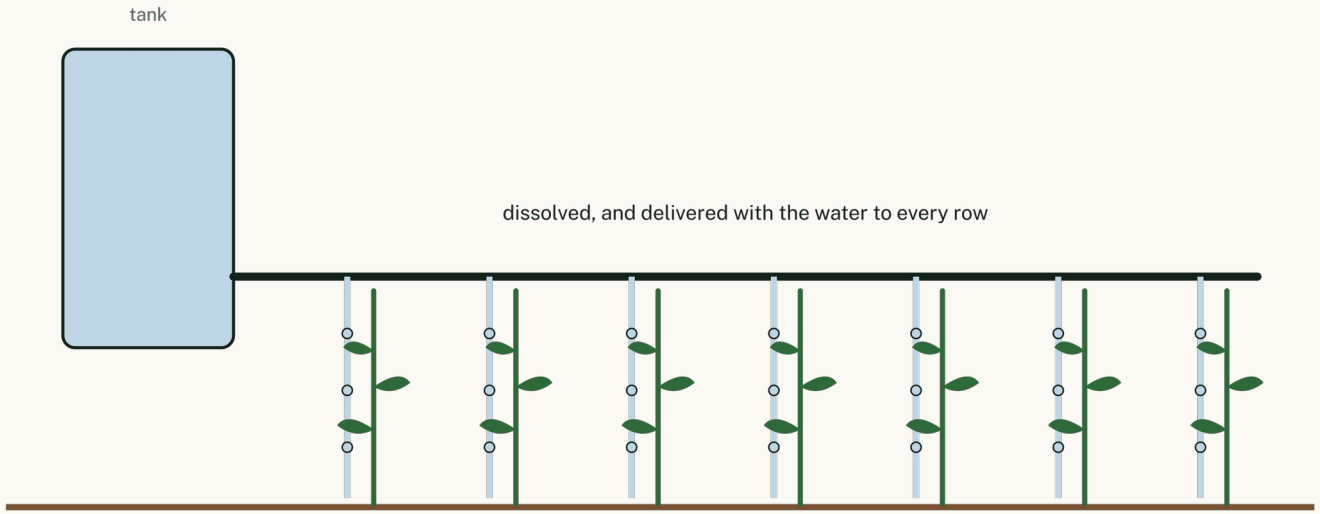
Crop	Humi Grow, 8 kg pellet	Humi Grow Plus, 4 L liquid	Stages	Rupees per acre, both
Wheat	1 bag, land preparation, drill at sowing (or broadcast)	1 pack, germination, fertigation	At or before sowing; germination	3,990
Maize	1 bag, land preparation, drill at sowing (or broadcast)	2 packs, germination, fertigation	At or before sowing; germination	5,230
Potato	1 bag, land preparation, drill at sowing (or broadcast)	2 packs, early growth, fertigation	At or before sowing; early growth	5,230
Garlic	1 bag, land preparation, drill at sowing (or broadcast)	1 pack, germination, fertigation	At or before sowing; germination	3,990
Citrus	2 bags, land preparation, broadcasting	1 pack, germination, fertigation	Before the season; germination	6,740

The crop plan adds the liquid to the pellet; it never replaces one with the other. The pellet is a bulk soil application at or before sowing; the liquid is placed in the wetted zone after the crop is in. The 2 differ in form, rate and stage, so do not compare them on the percentage of humic acid alone.

Rows as printed in VAN's published crop plans on van.com.pk. Humi Grow 8 kg Rs 2,750; Humi Grow Plus 4 L Rs 1,240 (the VAN farmer price sheet, August 2026). Wheat 2,750 + 1,240 = 3,990; maize 2,750 + 2,480 = 5,230; citrus 5,500 + 1,240 = 6,740.

HOW IT IS APPLIED

Through the water at the stage the crop plan names: into the drum, then released over the whole irrigation.



Measure it, then dilute it

1 pack of 4 L at germination on wheat, garlic and citrus; 2 packs on maize at germination and on potato at early growth. Pour it into a 200 L drum of water at the head of the watercourse, or into the injector tank of a drip system, and stir it in. The drawing shows a 200 L drum, the usual drum at the head of a watercourse. If your drum is a different size, change the amount of water, never the amount of product.

Meter it in, with the water already flowing

Open the tap so the drum empties over the whole irrigation, not all at once. The liquid goes where the water goes: into the wetted root zone across the whole acre. If it all goes in at the start, the first strip gets it and the last strip gets none.

At the stage your plan prints; never as a foliar spray

Every field-crop row applies it at germination or early growth, while the root system is forming, except the sugarcane ratoon, which takes it at grand growth. Strawberry takes it again at flowering. Every row says fertigation. No foliar rate is given, because no crop plan row names one.

OBJECTION 1 OF 8

“1,240 for 4 litres, and it is only 10%?”

What the dealer says

Write down the price, and then what the pack delivers: about 400 g of humic acid (reading the label as w/v), already dissolved, placed in the wetted root zone across the whole acre at germination, while the crop is standing. Wheat takes 1 pack, potato 2. The pellet is 40% humic acid and goes on dry at or before sowing, worked into the soil. The liquid is 10% and goes down the line to the roots while they are forming. They are different forms used at different stages.

The facts behind the answer

- 4 L, Rs 1,240, delivered (the farmer price sheet, August 2026).
- Wheat 1 pack; maize and potato 2 packs (the crop plans on van.com.pk).
- About 400 g of humic acid and 140 g of K_2O per pack, reading the label as w/v.
- VAN’s founder first made humic acid in Pakistan in 2007 (the company timeline on van.com.pk).

Never say: “It pays for itself in saved DAP” or any rupees-per-acre return. No percentage saved and no yield figure exists for the product.

OBJECTION 2 OF 8

“If it unlocks my phosphate, can I cut a bag of DAP?”

What the dealer says

No, and do not tell a grower that it makes more of his phosphate available. VAN checked whether humic acid at field rates can stop calcium from binding phosphate. It cannot: the calcium it binds is far too little. VAN therefore makes no such claim and gives no figure for phosphate saved. A humate holds nutrients on exchange sites, keeps micronutrients in solution and helps the root. Keep the phosphate rate as the crop plan prints it.

The facts behind the answer

- Humic acid benefits the soil and the root. van.com.pk does not claim that it stops calcium from binding phosphate.
- VAN’s 2026 calculation of how much calcium humic acid can bind: at any field rate, it binds too little calcium to stop calcium from binding phosphate.
- No percentage of phosphate saved is printed anywhere by VAN.

Never say: “It saves X% of your phosphate” or “cut the DAP by a bag”.

OBJECTION 3 OF 8

“I used it and my soil test did not change.”

What the dealer says

It will not, and VAN says so. 1 pack is about 400 g of humic acid on an acre of soil, and a soil test measures the whole acre to plough depth. The effect is in the soil around the young roots: it holds more of the nutrients you apply, keeps micronutrients in solution, and helps the root grow. A soil test will not show that effect, because it measures the whole acre to plough depth.

The facts behind the answer

- About 400 g of humic acid per pack (the declared analysis, read as w/v).
- Mean organic matter 0.61%; 97.6% of samples below 1% (the Punjab soil survey, 770,160 samples, about 2016 to 2018).
- If a soil test shows more phosphorus after humic acid is added, that alone does not prove the crop can use more phosphorus (general soil chemistry).

Never say: “Your organic matter will rise by X” or “the report will read higher next season”.

OBJECTION 4 OF 8

“Can I spray it on the leaf instead?”

What the dealer says

Every crop plan row for Humi Grow Plus says fertigation, and the label names fertigation as the method, so that is the only method given here. It is made to go into the soil solution where the roots are. Where the crop plan gives a foliar spray, it names other products, such as the VL liquids and Tornado. Do not give a foliar rate for this one.

The facts behind the answer

- The method on the label and the product page is fertigation; the crop plan rows: fertigation on every row (van.com.pk).
- The foliar rows on the same crop plans belong to the VL liquids and Tornado.
- No foliar rate for Humi Grow Plus is published anywhere.

Never say or give: any foliar rate, such as “1 L in 100 L of water”.

OBJECTION 5 OF 8

“Which stage, and how many packs?”

What the dealer says

The crop plans apply it at germination on wheat, garlic and citrus (1 pack), at germination on maize (2 packs), at early growth on potato (2 packs), at early growth and again at flowering on strawberry (2 packs each time), and at grand growth on the sugarcane ratoon (2 packs). In every case it goes through the water after the crop is sown. On field crops the crop plans apply it at germination or early growth, while the root system is forming. The exceptions are the sugarcane ratoon, at grand growth, and strawberry, which takes a second application at flowering.

The facts behind the answer

- Rates per acre as printed in the crop plans on van.com.pk; 7 crop plans on the crop plan slides; the stage is the one each crop plan gives.
- The root system forms in the first weeks; a humate placed then reaches the root while it is forming.

Never say: “3 packs for a bad soil.” Use the crop plan rate; VAN has no figure for a higher dose.

OBJECTION 6 OF 8

“It has potash in it. Do I count it?”

What the dealer says

Count it, but do not rely on it. K_2O 3.5% of 4 L is about 140 g a pack. The potash in the crop plan comes from its potash rows: Fusion Potash at land preparation, Vital Potash through the water, and on some plans VL-Potash Liquid sprays (wheat) or Crop Force. The 140 g in this pack does not change any of them. Record it with the other potash applied.

The facts behind the answer

- About 140 g of K_2O per pack: 4 L × 3.5% w/v (the declared analysis).
- The potash rows on the same crop plans are Fusion Potash at land preparation, Vital Potash through the water, and on some plans VL-Potash Liquid sprays (wheat) or Crop Force (the crop plans on van.com.pk).
- Too much potash reduces the uptake of magnesium and calcium; at 140 g this pack is too small to have that effect.

Never say: “It is a potash fertiliser too” or that it replaces any part of the potash on the crop plan.

OBJECTION 7 OF 8

“Is it registered? What is the licence?”

What the dealer says

It is registered under Pakistan Standard PS 5610-2023, with manufacturing licence CM/L-3274/2024, and VAN is the registered maker. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. Enter the batch number from the pack on van.com.pk, and you can read the certificate for that batch.

The facts behind the answer

- Humi Grow Plus: PS 5610-2023, CM/L-3274/2024 (the regulatory table on van.com.pk).
- Humi Grow, the pellet, carries the same standard under CM/L-3281/2024.
- Every batch is released through VAN Lab, PNAC LAB 336, ISO/IEC 17025:2017.

Never say or give: a licence expiry date, or the licence as a claim about performance.

OBJECTION 8 OF 8

“I already put the Humi Grow pellet on at land preparation. Why the liquid too?”

What the dealer says

Because the crop plan uses them at 2 different stages, for 2 different jobs. The pellet went in at or before sowing, drilled or worked into the top layer of soil across the field. The liquid goes in at germination or early growth, with the crop standing, and reaches the wetted zone where the young roots are, which the drill and the spreader cannot reach once the crop is in. Wheat takes both: 1 bag of the pellet, then 1 pack of the liquid, Rs 3,990 per acre for the 2. Neither replaces the other.

The facts behind the answer

- All 7 crop plans that name Humi Grow Plus also use Humi Grow; for example, wheat, maize, potato, garlic and citrus carry Humi Grow at land preparation and Humi Grow Plus at germination or early growth (the crop plans on van.com.pk).
- Humi Grow 8 kg Rs 2,750 and Humi Grow Plus 4 L Rs 1,240: Rs 3,990 on wheat (the farmer price sheet, August 2026).
- The same active ingredient in 2 forms, used at 2 stages.

Never say: “Skip the pellet if you use the liquid”, or the reverse, or “the 2 together give X”.

ITS PLACE IN THE CROP PLANS

Humi Grow Plus in the published crop plans, per acre.

Crop	Stage	Rate per acre	Method	Pack
Wheat	Germination	1 pack	Fertigation	4 L
Maize (Corn)	Germination	2 packs	Fertigation	4 L
Ratoon Sugarcane	Grand Growth	2 packs	Fertigation	4 L
Potato	Early Growth	2 packs	Fertigation	4 L
Garlic	Germination	1 pack	Fertigation	4 L
Citrus, 5 years and over	Germination	1 pack	Fertigation	4 L
Strawberry	Early Growth	2 packs	Fertigation	4 L
Strawberry	Flowering	2 packs	Fertigation	4 L

The rows are printed as published in the crop plans on van.com.pk. Coverage: these are all 8 rows in the 7 crop plans that name Humi Grow Plus. Every row says fertigation and uses the 4 L pack. Each rate is per acre, and the crop page scales it to the size of the field.

WHAT VAN DOES NOT CLAIM FOR THIS PRODUCT

A dealer who makes any of these claims is speaking for himself, not for VAN.

1. A yield figure, or a nutrient use efficiency percentage. No Humi Grow Plus trial exists. Describe the 81 published studies as research on humic substances in general, never as a result for this product.

2. That it stops calcium from binding phosphate, or saves any percentage of phosphate or nitrogen. VAN's own check rules out the mechanism, and van.com.pk gives no figure for phosphate saved.

3. A soil organic-matter or soil-test change. About 400 g of humic acid on an acre acts in the root zone; it does not change the soil test.

4. A foliar rate. Every crop plan row and the label say fertigation.

5. That it replaces the pellet, or the pellet replaces it, or that the 2 together give a measured result.

6. A 20 L price, or that it is the main source of potash. Only the 4 L pack is on the VAN farmer price sheet, August 2026, and its K_2O is 140 g.

HOW A GROWER VERIFIES A PACK

Every batch is tested, and the batch number on the pack identifies its certificate.

1 · The batch number

Every pack carries the batch number. Read it to the grower and show him where it is printed.

2 · The certificate of analysis

On van.com.pk, a grower can read the certificate of analysis for any pack by entering the batch number printed on it. The grower does not need to post a sample.

3 · The laboratory

VAN Lab is accredited by the Pakistan National Accreditation Council (PNAC) to ISO/IEC 17025:2017, the international standard for testing laboratories, as LAB 336. Every batch is released through the laboratory before it ships. The certificate carries the laboratory's number.

What a certificate for Humi Grow Plus shows

The certificate shows the tested humic acid and K_2O content of the batch against the declared 10% and 3.5%. It also shows the batch number, and the Pakistan Standard PS 5610-2023 and licence CM/L-3274/2024 printed on the pack. This training shows no batch result; a grower can read the certificate of his own batch on van.com.pk by its batch number. Commercial humic products elsewhere can differ widely from their labels, which is why the grower should read the certificate for his batch.

THE URDU LINES · REVIEW STATUS UNDER EACH LINE

The lines a dealer says in Urdu.

The first line is the product's main message. Where its nutrient has one, the line gives VAN's promise word: the 1-word brand message VAN uses for that nutrient. The other 3 lines are key sentences of this deck. Each English line gives the meaning of the Urdu beside it. Send any correction to partner@van.com.pk.

A humate helps the soil hold the fertiliser you apply. This one goes down the water line. *English meaning · Urdu reviewed 27 Sep 2026*	ہیومک زمین کو وہ گرفت دیتا ہے جس سے وہ کھاد کو تھامے رکھے، اور یہ پانی کی لائن سے جاتا ہے
Put it in the drum at germination or early growth, with the water flowing. *English meaning · Urdu reviewed 27 Sep 2026*	اُگاؤ یا ابتدائی بڑھوتری پر ڈرم میں ڈالیں، پانی چلتا رہے
It helps the root and the soil. It does not change your soil test, and VAN says so. *English meaning · Urdu reviewed 27 Sep 2026*	یہ جڑ اور زمین کی مدد کرتا ہے۔ یہ آپ کی مٹی کی رپورٹ نہیں بدلتا، اور VAN یہ کھل کر کہتا ہے
The pellet goes in at or before sowing; the liquid goes in with the crop standing. *English meaning · Urdu line pending review*	دانہ بوائی پر یا اس سے پہلے، مائع کھڑی فصل کے ساتھ

TALK TO VAN

Use the address that matches your request.

For growers

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch or a dealer near you.

For the trade

partner@van.com.pk

Dealers, distributors and the trade: stock, training and corrections to this deck.

For everything else

info@van.com.pk

Supply, the laboratory or a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore, has 1 helpline and 3 addresses. VAN sends no group messages; every grower who writes gets a direct reply.

Humi Grow Plus puts humate into the water at germination and early growth; it does not stop calcium from binding phosphate.

